BE TRUE

TRINITY COLLEGE NABBINGO A LEVEL CHEMISTRY SEMINAR, 26th September, 2026

ANSWER GUIDE

NUMBER 1

ITEM 1

Electrochemical cell

$E^{\ominus}_{cathode} = +0.34V$
$E^{\ominus}_{anode} = -0.76V$
Cell potential:

$E^{\ominus}_{cell} = E^{\ominus}_{cathode} - E^{\ominus}_{anode}$ ✓L
$= (+0.34) - (-0.76)$
$\boxed{E^{\ominus}_{cell} = +1.10V}$ ✓Da

Since the monitoring equipment requires at least 1.00 V, and the proposed cell provides 1.10 V under standard conditions. Therefore, based strictly on the stated standard-condition requirement, the cell provides a potential difference 0.10 V above the minimum requirement and is suitable for supplying the required potential difference.✓DI

Gibbs free energy change:

$\Delta G^{\ominus} = -nFE^{\ominus}_{cell}$ ✓L
$= -2 \times 96500 \times 1.10$
$\boxed{\Delta G^{\ominus} = -212300J\ mol^{-1}}$
$\boxed{\Delta G^{\ominus} = -212.3kJ\ mol^{-1}}$ ✓Da

(ii) Copper recovery by electrolysis

$I = 19.3A$
$t = 2.00 \times 60 \times 60 = 7200s$
$Q = It$ ✓L
$= 19.3 \times 7200$
$\boxed{Q = 138960C}$ ✓Da
$n(e^-) = \frac{Q}{F}$

$= \frac{138960}{96500}$
$= 1.44mol$ ✓Da

Therefore, 1.44 moles of electrons passed through the electrolytic cell.

From: $Cu^{2+} + 2e^- \rightarrow Cu$

*2 moles of electrons deposit 1 mole of Cu*

$n(Cu) = \frac{1.44}{2} = 0.720mol$
$m = nM$
$= 0.720 \times 63.5$
$\boxed{m = 45.7g}$ ✓Da

The company requires at least **22.0 g of copper per cycle**, whereas the calculated deposition is approximately **45.7 g per cycle**. The stated operating conditions provide **more than twice the required copper output**, assuming the process operates with 100% current efficiency.✓DI

**Effect of the operating factors on copper recovery**



Current supplied

Increasing the current increases the amount of copper deposited per unit time, provided other conditions remain suitable. A decrease in current reduces the deposition rate.✓Fe

Duration of electrolysis

Increasing the electrolysis time increases the quantity of copper deposited because more charge passes through the electrolyte.✓Fe

Concentration of $Cu^{2+}$ ions

Increasing the concentration of $Cu^{2+}$ provides more copper ions available for reduction at the cathode and can support efficient copper deposition. If the $Cu^{2+}$ concentration becomes too low, copper deposition can become less efficient, and other electrode processes may become significant.✓Fe

b) Thermochemical evaluation

Reaction:

$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$
$\Delta H^{\ominus} = \sum \Delta H_f^{\ominus}(products) - \sum \Delta H_f^{\ominus}(reactants)$ ✓L
$= [0 + (-286)] - [(-155) + 0]$
$= -286 + 155$
$\boxed{\Delta H^{\ominus} = -131kJ\ mol^{-1}}$ ✓Da

Since $\Delta H^{\ominus}$ is negative, the reaction is **exothermic**.✓DI

The heat released can help provide the heat required to sustain the reduction process, thereby **reducing the amount of external energy required for heating**.✓Ae

**Recommendation**

The company should consider adopting the hydrogen reduction process as an alternative for producing copper from CuO because it is exothermic and can reduce external heating requirements.✓r

ITEM 2

Battery system 1: Reaction: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
Given:

$E^{\ominus}_{Cu^{2+}/Cu} = +0.34V$
$E^{\ominus}_{Zn^{2+}/Zn} = -0.76V$
$E^{\ominus}_{cell} = E^{\ominus}_{cathode} - E^{\ominus}_{anode}$ ✓L
$= (+0.34) - (-0.76)$
$\boxed{E^{\ominus}_{cell} = -1.10V}$ ✓Da

The reaction involves the transfer of 2 electrons.

$\Delta G^{\ominus} = -nFE^{\ominus}_{cell}$ ✓L
$= -2 \times 96500 \times 1.10$
$\boxed{\Delta G^{\ominus} = -212.3\ kJ\ mol^{-1}}$ ✓Da



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The battery is required to release a minimum of 300.0 kJ $mol^{-1}$ of electrical energy, but battery system 1 provides only 212.3 kJ mol. Therefore, **battery system 1 is not suitable**, because it does not provide the minimum electrical energy required to keep the radio transmitter operating.✓Di

**Battery system 2**

$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$
$E^\ominus_{cell} = (+0.80) - (-0.76)$
$\boxed{E^\ominus_{cell} = +1.56V}$ ✓Da

$\Delta G^\ominus = -nFE^\ominus_{cell}$
$= -2 \times 96500 \times 1.56$
$\boxed{\Delta G^\ominus = -301.1\ kJ\ mol^{-1}}$ ✓Da

Two electrons are transferred: $n = 2$

The battery is required to release at least 300.0 kJ $mol^{-1}$, while battery system 2 releases 301.1 kJ $mol^{-1}$. Therefore, **battery system 2 is suitable**, since it meets the minimum electrical energy requirement.✓Di

**Heater**

Required reaction: $\boxed{C(s) - 2H_2(g) \rightarrow CH_4(g)}$

**Step 1: Write and number the given equations**

**(Eq1)** $C(s) - O_2(g) \rightarrow CO_2(g)$
$\Delta H_1 = -393.5\ kJ\ mol^{-1}$

**(Eq2)** $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$

Since the required reaction contains **2 mol of** $H_2$, multiply equation (2) by 2:

**(Eq3) = 2 × (2)**

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
$\Delta H_3 = 2(-285.8)$
$\boxed{\Delta H_3 = -571.6\ kJ\ mol^{-1}}$

**(Eq4)** $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ :
$\Delta H_4 = -890.3\ kJ\ mol^{-1}$

Reverse equation (4): **(Eq5) = -(Eq4)**

$\underline{CO_2(g) + 2H_2O(l) \rightarrow CH_4(g) + 2O_2(g)}$
$\boxed{\Delta H_5 = +890.3\ kJ\ mol^{-1}}$

**Step 2: Add equations (1), (3), and (5)**

$C + O_2 \rightarrow CO_2$
$2H_2 + O_2 \rightarrow 2H_2O$
$CO_2 + 2H_2O \rightarrow CH_4 + 2O_2$

Cancel:

- $CO_2, 2H_2O, 2O_2$ remaining:

$\boxed{C + 2H_2 \rightarrow CH_4}$

**Step 3:** Add the enthalpy changes: $\Delta H =$
$\Delta H_1 + \Delta H_3 - \Delta H_5$
$= (-393.5) + (-571.6) + (+890.3)$
$\boxed{\Delta H = -74.8\ kJ\ mol^{-1}}$ ✓Da

For 2.0 mol of carbon:

$Q = 2.0 \times -74.8$
$\boxed{Q = -149.6\ kJ}$ ✓Da

The heater is required to release at least 140.0 kJ, while the reaction involving the supplied 2.0 mol of carbon releases 149.6 kJ. Therefore, **the heater provides sufficient heat to melt the ice and restore the electrolyte to a functional state.**✓Di

**(b) Factors favouring sufficient heat production**

- **Amount of carbon and hydrogen:** Increasing the amount of reactants increases the total amount of heat released because more substance reacts. ✓Fe



- **Suitable temperature and pressure:** Maintaining appropriate conditions allows the reaction to proceed effectively and at a sufficient rate to generate heat when required. ✓Fe

**Other applications of the chemical systems**

**1. Electrochemical cells — batteries**

Electrochemical cells convert **chemical energy into electrical energy** through spontaneous redox reactions. They are therefore used in **batteries** to supply electrical power to devices such as radios, phones, vehicles and emergency equipment.✓Ae

**2. Electrolysis**

Electrolysis uses electrical energy to drive a **non-spontaneous chemical reaction**. It is used in **electroplating**, where a thin layer of one metal is deposited on another object to improve its appearance or protect it from corrosion.✓Ae

It is also used in the **extraction and purification of metals**. For example, electrolysis can be used to obtain or purify copper for use in electrical wiring.✓Ae

**3. Methane combustion**

Methane undergoes combustion and releases a large amount of heat. The heat released is used for **cooking, heating buildings, and industrial heating processes**. Methane can also be used as a fuel in power generation, where its chemical energy is converted into electrical energy.✓Ae

## NUMBER 2 SOLUTIONS

**ITEM 1**

**Determination of the molecular formula of HA**

The general formula of a saturated monocarboxylic acid is:

$$C_nH_{2n}O_2$$

The given data are:

- Volume of HA vapour = **10.0 $cm^3$**
- Volume of oxygen initially = **70.0 $cm^3$**
- Volume after combustion and cooling = **50.0 $cm^3$**
- Volume after passing through aqueous KOH = **30.0 $cm^3$**

**Step 1: Determine the volume of carbon dioxide produced**

Aqueous KOH absorbs carbon dioxide. $V_{CO_2} = 50.0 - 30.0$



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$V_{CO_2} = 20.0\ cm^3$

Since all gases are measured at the same temperature and pressure, their volumes are proportional to their number of moles.

Therefore,

$$\frac{V_{CO_2}}{V_{HA}} = \frac{n}{1}$$
$$\frac{20.0}{10.0} = n$$
$$n = 2$$

Therefore, the molecular formula of HA is:

$$C_2H_4O_2$$

Hence, HA is ethanoic acid, $CH_3COOH$.

**2. Determination of the concentration of HA**

The neutralisation reaction is:

$$HA(aq) + NaOH(aq) \rightarrow NaA(aq) + H_2O(l)$$

1000 $cm^3$ of solution contains 0.1 moles.

20.0 $cm^3$ of solution contains: $\left(\frac{0.1 \times 20.0}{1000}\right)$ ✓Da
= 0.002 moles

**Moles of HA**

According to the mole ratio:

1 mole of HA reacts with 1 mole of NaOH.

∴ Moles of HA = 0.002 moles.

25.0 $cm^3$ of HA contains 0.002 moles.

1000 $cm^3$ of HA contains: $\left(\frac{0.002 \times 1000}{25}\right)$ ✓Da
= 0.08 M

1000 $cm^3$ = 1 $dm^3$

1 $dm^3$ contains 0.08 moles.

500 $dm^3$ contains: $\left(\frac{0.08 \times 500}{1}\right)$ ✓Da
= 40 moles

**Molar Mass of HA**

The acid was $C_2H_4O_2$

$RMM = (12 \times 2) + 4 + 32$
= 60.0 g

**Moles = Mass / RMM**

Mass = Moles × $RMM$
= 40 × 60
= 2400 g ✓Da

Therefore, the 500 $dm^3$ production batch contains 40.0 moles of ethanoic acid, corresponding to 2400 g of HA. ✓Di

**b) Evaluate the proposed buffer formulation**

The proposed buffer contains: HA = 0.100 mol $dm^{-3}$, NaA = 0.080 mol $dm^{-3}$ $K_a = 1.8 \times 10^{-5}$

- Required pH = 4.60–4.80

The buffer consists of a weak acid, HA, and its conjugate base, $A^-$.

The ionisation equilibrium is:

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

The acid dissociation constant is:

$$K_a = \frac{[H^+][A^-]}{[HA]}$$



$pK_a = -\log K_a$ ✓L
$= -\log(1.8 \times 10^{-5})$
$pK_a = 4.74$ ✓Da

**Calculate the pH of the proposed buffer**

Using the Henderson–Hasselbalch equation:

$$pH = pK_a + \log\frac{[A^-]}{[HA]}$$ ✓L

Since NaA provides $A^-$: $[A^-] = 0.080$ mol $dm^{-3}$
and: $[HA] = 0.100$ mol $dm^{-3}$
Therefore:

$pH = 4.74 + \log\left(\frac{0.080}{0.100}\right)$ ✓Da
$pH = 4.74 + \log(0.80)$

$\log(0.80) = -0.0969$
Therefore:

$pH = 4.74 - 0.0969$
$pH = 4.65$ ✓Da

**Interpretation of the calculated pH**

The company requires a pH range of: 4.60 - 4.80
The calculated buffer pH is: 4.65

Since:

$$4.60 < 4.65 < 4.80$$

the calculated pH falls within the required range.

Therefore:

The proposed buffer formulation is suitable with respect to pH. ✓Di

**How the buffer resists changes in pH**

A buffer contains a **weak acid (HA)** and its **conjugate base ($A^-$)**.

- When acid ($H^+$) is added, the conjugate base removes it: $A^- + H^+ \rightarrow HA$ ✓Fe
  This prevents a large decrease in pH.
- When alkali ($OH^-$) is added, the weak acid reacts with it: $HA + OH^- \rightarrow A^- + H_2O$ ✓Fe
  This prevents a large increase in pH.

Therefore, the presence of appreciable amounts of both HA and $A^-$ allows the buffer to resist sudden changes in pH.

**Application in fruit processing**

The buffer can be used to maintain the fruit product within the desired pH range of 4.60–4.80 despite small amounts of acid or alkali entering during processing or storage. This helps maintain consistent taste, product stability, and shelf life. ✓Ae

**ITEM 2**

The buffer consists of a weak acid, $CH_3COOH$, and its conjugate base, $CH_3COO^-$.

Using the Henderson–Hasselbalch equation:

$$pH = pK_a + \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$ ✓L

First calculate the number of moles.

Molar mass of $CH_3COOH$ = 60 g $mol^{-1}$



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$n(CH_3COOH) = \frac{4.50}{60} = 0.0750\ mol$

Molar mass of $CH_3COONa$ = 82 g mol$^{-1}$

$n(CH_3COONa) = \frac{2.05}{82} = 0.0250\ mol$

Since both are dissolved in the same volume, the volume cancels when calculating their concentration ratio:

$\frac{[CH_3COO^-]}{[CH_3COOH]} = \frac{0.0250}{0.0750} = \frac{1}{3}$

Calculate $pK_a$:

$pK_a = -\log(1.8 \times 10^{-5})$

$pK_a = 4.74$

Therefore,

$pH = 4.74 - \log\left(\frac{1}{3}\right)$

$pH = 4.74 - 0.477$

$\boxed{pH_Q = 4.26}$ ✓Da

This is below the required range of 4.8–5.3.

Therefore, formulation Q would not provide the required pH for maximum stability of the antibiotic.

**Step 2: Calculate the pH of formulation R**

Moles of $CH_3COOH$:

$n = \frac{1.50}{60} = 0.0250\ mol$

Moles of $CH_3COONa$:

$n = \frac{6.15}{82} = 0.0750\ mol$

Therefore,

$\frac{[CH_3COO^-]}{[CH_3COOH]} = \frac{0.0750}{0.0250} = 3$ ✓Da

Using:

$pH = pK_a + \log\frac{[CH_3COO^-]}{[CH_3COOH]}$

$pH = 4.74 + \log 3$

$pH = 4.74 + 0.477$

$\boxed{pH_R = 5.22}$ ✓Da

The required pH range is:

4.8 – 5.3

Since: 4.8 < 5.22 < 5.3 formulation R lies within the required range.

**Recommendation**

Formulation R is the more suitable formulation.

It provides a pH of approximately 5.22, which falls within the required range of 4.8–5.3. Therefore, it should better maintain the stability and therapeutic effectiveness of the antibiotic. ✓Di

**Analysis of precipitation in batches A, B and C**

Calcium ions react with carbonate ions according to:

$Ca^{2+}(aq) + CO_3^{2-}(aq) \rightleftharpoons CaCO_3(s)$

The solubility product expression is:

$\boxed{K_{sp} = [Ca^{2+}][CO_3^{2-}]}$ ✓L

Given:

$K_{sp} = 4.8 \times 10^{-9}$

We calculate the ionic product, $Q_{sp}$ for each batch.

**Batch A**

Given:

$[Ca^{2+}] = 1.0 \times 10^{-3}$ $[CO_3^{2-}] = 2.0 \times 10^{-6}$

Therefore:

$Q_{sp} = (1.0 \times 10^{-3})(2.0 \times 10^{-6})$

$\boxed{Q_{sp} = 2.0 \times 10^{-9}}$ ✓Da

Compare:

$Q_{sp} < K_{sp}$

$2.0 \times 10^{-9} < 4.8 \times 10^{-9}$

Therefore, the solution is unsaturated with respect to $CaCO_3$. ✓Di

No $CaCO_3$ precipitate is expected in batch A.

**Batch B**

$Q_{sp} = (2.5 \times 10^{-3})(3.0 \times 10^{-6})$

$\boxed{Q_{sp} = 7.5 \times 10^{-9}}$ ✓Da

Compare:

$Q_{sp} > K_{sp}$

$7.5 \times 10^{-9} > 4.8 \times 10^{-9}$

Therefore, the solution is supersaturated. ✓Di

$CaCO_3$ precipitation is expected in batch B.

**Batch C**

$Q_{sp} = (4.0 \times 10^{-3})(2.0 \times 10^{-5})$

$\boxed{Q_{sp} = 8.0 \times 10^{-8}}$ ✓Da

Compare:

$Q_{sp} > K_{sp}$

$8.0 \times 10^{-8} > 4.8 \times 10^{-9}$

Therefore, batch C is highly supersaturated and substantial precipitation is expected. ✓Di

**Maintains drug stability:**

The buffer keeps the antibiotic within the required pH range of 4.8–5.3, reducing decomposition of the active ingredient and preserving its therapeutic effectiveness. ✓Ae

**Resists sudden pH changes:**

The ethanoic acid/ethanoate buffer reacts with small amounts of added acid or alkali, preventing large changes in pH during manufacturing, storage and use. ✓Ae

**Ensures consistent drug effectiveness:**

Maintaining the correct pH ensures that each dose contains the antibiotic in a stable form and therefore provides a reliable therapeutic effect. ✓Ae

**Protects patient safety:**

A properly controlled formulation reduces the risk of degradation products or precipitates forming in an intravenous preparation, helping to ensure safe administration to patients. ✓Ae

**Improves shelf life:**

By controlling pH and preventing conditions that accelerate decomposition, the buffer helps the medicine remain effective for longer during storage. ✓Ae

Precipitation can be minimised by reducing the concentrations of $Ca^{2+}$ and $CO_3^{2-}$, ✓Fe using purified materials, preventing carbonate contamination, controlling pH and monitoring the ionic product. The ionic product should be maintained below $K_{sp}$, while the buffer keeps the antibiotic within its optimum





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pH range of 4.8-5.3. This prevents $CaCO_3$ precipitation without compromising drug stability and effectiveness.✓Fe

## ITEM 3

### (a) Ionic Product and Solubility Product

The dissolution equilibrium of calcium carbonate is:

$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$

**Sample A**

Ionic product $= [Ca^{2+}][CO_3^{2-}]$ ✓L

$= (1.0 \times 10^{-2})(2.0 \times 10^{-7})$

$= 2 \times 10^{-9}\ mol^2\ dm^{-6}$ ✓Da

**Sample B**

Ionic product $= (2.5 \times 10^{-2})(3.0 \times 10^{-7})$

$= 7.5 \times 10^{-9}\ mol^2\ dm^{-6}$ ✓Da

**Sample C**

Ionic product $= (4.0 \times 10^{-2})(5.0 \times 10^{-7})$

$= 2 \times 10^{-8}\ mol^2\ dm^{-6}$ ✓Da

Sample B and sample C are more likely to form visible deposits since their ionic product values are greater than the solubility product. This means that their solutions are supersaturated, and therefore the excess ions combine to precipitate calcium carbonate.✓Di

### (b) Solvent Extraction

Let the mass of Q extracted be $y$.

$K_D = \frac{[Q]_{organic\ solvent}}{[Q]_{water}}$ ✓L

$4.0 = \frac{y/25}{(0.8-y)/100}$

$4 = \frac{100y}{(0.8-y)25}$

$4(0.8-y) = 4y$

$3.2 - 4y = 4y$

$8y = 3.2$

$y = 0.4\ g$ ✓Da

**Mass remaining in the solution** $= 0.8 - 0.4$

$= 0.4\ g$ ✓Da

**For the second successive extraction**

$4 = \frac{y/25}{(0.4-y)/100}$

$4 = \frac{100y}{(0.4-y)25}$

$4(0.4-y) = 4y$

$1.6 - 4y = 4y$

$8y = 1.6$



$y = 0.2\ g$ ✓Da

**Total mass extracted** $= 0.4 + 0.2$

$= 0.6\ g$ ✓Da

**Percentage of Q extracted**

Percentage extracted $= \frac{0.6 \times 100}{0.8}$

$= 75\%$ ✓Da

The purification method used is effective because it extracts a higher percentage of the contaminant Q than what remains in the water.✓Di

### (c) Factors Affecting Precipitation of Calcium Carbonate

The water picked from underground may at times contain other dissolved salts having at least one ion similar to the ions from calcium carbonate. This would increase the concentration of the ions so that the ionic product of calcium and carbonate ions would exceed the solubility product of calcium carbonate. Hence, the excess ions would react to precipitate out calcium carbonate in such water samples so as to restore equilibrium and the solubility product value of calcium carbonate.✓Fe

Some water samples picked from underground might contain compounds that form complexes with calcium ions (complexing agents). This would reduce the concentration of the calcium ions which are free in the water and therefore more of the calcium carbonate present in water would dissolve. No precipitation would therefore occur since the salt would be becoming more soluble.✓Fe

In water samples stored under high temperatures, precipitation cannot easily occur. This is because the solubility product of calcium carbonate is high at high temperatures and equilibrium requires high concentrations of calcium and carbonate ions to be established, which are not present in the water sample. Therefore, the ionic product of calcium and carbonate ions will be less than the solubility product value of calcium carbonate, meaning the solution will be dilute at that temperature; hence, no precipitation occurs.✓Fe

**Applications of Ionic Equilibria**

**1. Manufacture of Soap**

Ionic equilibria can be applied in the manufacture of soap. A concentrated solution of sodium chloride is added to the soap solution and this greatly increases the concentration of sodium ions since soap also possesses sodium ions. The ionic product therefore exceeds the solubility product value of soap, so that the excess sodium ions combine with carboxylate ions to precipitate out soap from its solution.✓Ae

**2. Manufacture of Steel**



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The knowledge of eutectic mixtures from physical equilibria can be used to make steel, an alloy of carbon and iron. A liquid mixture containing carbon and iron is cooled until the whole liquid mixture solidifies at a constant temperature to form solid carbon and solid iron, which is steel.✓Ae

## ITEM 4

## NUMBER 3/4 SOLUTIONS

ITEM 1

**Graphite**

Graphite has a **giant covalent layered structure** in which each carbon atom is strongly covalently bonded to three other carbon atoms. It has a very high sublimation temperature because a large amount of energy is required to overcome the strong covalent bonds. Graphite conducts electricity in the solid state because each carbon atom has one delocalised electron that is free to move through the layers and carry electrical charge.✓Pe

**Silicon dioxide**

Silicon dioxide has a **giant three-dimensional covalent structure** in which silicon and oxygen atoms are joined by strong covalent bonds. It has a high melting point because considerable energy is required to break these strong bonds. It does not conduct electricity because there are no mobile ions or delocalised electrons available to carry electrical charge.✓Pe

**Sodium oxide**

Sodium oxide has a **giant ionic structure** containing positively charged sodium ions and negatively charged oxide ions held together by strong electrostatic forces. It has a relatively high melting point because considerable energy is needed to overcome these attractions. It does not conduct electricity when solid because the ions are fixed in position, but it conducts when molten because the ions become free to move and carry charge.✓Pe

**(b) Chemical behaviour during processing and cleaning**

**1. Graphite with dilute hydrochloric acid**

Graphite does not react with dilute hydrochloric acid under ordinary conditions because graphite has a strong giant covalent structure and is relatively chemically unreactive. Graphite also does not react with water under ordinary conditions because its strong carbon-carbon covalent bonds are not readily broken by water.✓Re

**2. Silicon dioxide with dilute hydrochloric acid**



Silicon dioxide is an acidic oxide and therefore does not normally react with another acid. Silicon dioxide also does not react appreciably with water because its giant covalent structure is very stable and its silicon-oxygen bonds are not easily broken by water.

However, silicon dioxide reacts with hot concentrated sodium hydroxide because silicon dioxide is an acidic oxide whereas sodium hydroxide is an alkali. They therefore undergo an acid-base reaction to form sodium silicate and water.✓Re

$$SiO_2(s) + 2NaOH(aq) \rightarrow Na_2SiO_3(aq) + H_2O(l)$$ ✓eq

**3. Sodium oxide with water**

Sodium oxide reacts vigorously with water because it is a basic ionic oxide. The oxide ions react with hydrogen ions from water to produce hydroxide ions, forming sodium hydroxide.✓Re

$$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$$ ✓eq

**Sodium oxide with dilute hydrochloric acid**

Sodium oxide reacts with dilute hydrochloric acid because sodium oxide is a basic oxide while hydrochloric acid is an acid. The oxide ions react with hydrogen ions to form water, while sodium ions and chloride ions remain as sodium chloride.✓Re

$$Na_2O(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l)$$ ✓eq

**(c) Applications, health and environmental implications**

**Graphite**

Graphite is suitable for electrodes because it conducts electricity and for high-temperature equipment because of its very high sublimation temperature✓Ea. Fine graphite dust may irritate the eyes and respiratory system when inhaled.✓Ed the company should therefore control dust using good ventilation, dust extraction and appropriate protective equipment.✓Em

**Silicon dioxide**

Silicon dioxide is useful in the manufacture of glass, ceramics, concrete and refractory materials because of its high melting point and chemical stability✓Ea. However, prolonged inhalation of fine crystalline silica dust can cause serious lung disease such as silicosis✓Ed. The company should therefore use enclosed processes, effective ventilation, dust-suppression methods and appropriate respiratory protection.✓Em

**Sodium oxide**

Sodium oxide is mainly useful in the manufacture of glass and ceramics✓Ea. It is hazardous because it reacts with moisture to form corrosive sodium hydroxide. If released into water or moist soil, it



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can raise the pH and harm aquatic organisms✓Ed. It should therefore be stored in tightly sealed, moisture-proof containers and handled using suitable gloves, goggles and protective clothing.✓Em

ITEM 2

**Aluminium oxide, $Al_2O_3$**

Aluminium oxide has a **giant ionic lattice** containing aluminium ions and oxide ions held together by strong electrostatic attractions. It is a solid at room temperature and has a very high melting point of **2072 °C** because a large amount of energy is required to overcome the strong attractions between the oppositely charged ions. In the solid state, the ions are fixed in position, so aluminium oxide does not conduct electricity as a solid, although it conducts when molten because the ions become mobile.✓Pe

**Chlorine, $Cl_2$**

Chlorine has a **simple molecular structure** in which two chlorine atoms are joined by a strong covalent bond. However, the chlorine molecules are held together by **weak** London dispersion forces. Therefore, only a small amount of energy is required to separate the molecules, giving chlorine a very low melting point of **−101.5 °C**. It is consequently a gas at room temperature. Chlorine does not conduct electricity because it has no mobile ions or delocalised electrons.✓Pe

**Carbon dioxide, $CO_2$**

Carbon dioxide has a **simple molecular structure** in which each carbon atom is covalently bonded to two oxygen atoms. The carbon dioxide molecules are held together by weak intermolecular forces. Therefore, little energy is required to separate the molecules, giving carbon dioxide a very low sublimation temperature of **−78.5 °C**. It is consequently a gas at room temperature and does not conduct electricity because it contains no mobile ions or delocalised electrons.✓Pe

**(b) Explain their chemical behaviour with water and sodium hydroxide.**

**1. Aluminium oxide**

Aluminium oxide does not react appreciably with water under ordinary conditions because it has a stable giant ionic structure and is essentially insoluble in water.

Aluminium oxide reacts with sodium hydroxide because it is **amphoteric**, meaning that it can react with both acids and bases. In the presence of sodium hydroxide, aluminium oxide reacts with hydroxide ions to form a soluble hydroxoaluminate compound.✓Re

$$Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2Na[Al(OH)_4](aq)$$ ✓eq

**2. Chlorine**



Chlorine reacts with water because it undergoes **disproportionation**, whereby some chlorine is reduced while some is oxidised. The reaction produces hydrochloric acid and hypochlorous acid.✓Re

$$Cl_2(g) + H_2O(l) \rightarrow HCl(aq) + HClO(aq)$$ ✓eq

With cold, dilute sodium hydroxide, chlorine undergoes d sproportionation to form sodium chloride and sodium hypochlorite.✓Re

$$Cl_2(g) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$$ ✓eq

With hot, concentrated sodium hydroxide, chlorine forms sodium chloride and sodium chlorate.✓Re

$$3Cl_2(g) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$$ ✓eq

**3. Carbon dioxide**

Carbon dioxide reacts with water to form carbonic acid because carbon dioxide is an acidic oxide. The reaction is reversible because carbonic acid is a weak acid.✓Re

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$ ✓eq

The carbonic acid formed partially ionises in water. This explains why water containing dissolved carbon dioxide becomes weakly acidic.✓Re

$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$ ✓eq

Carbon dioxide reacts with sodium hydroxide because carbon dioxide is an acidic oxide while sodium hydroxide is an alkali. When sufficient sodium hydroxide is present, sodium carbonate and water are formed.✓Re

$$CO_2(g) + 2NaOH(aq) \rightarrow Na_2CO_3(aq) + H_2O(l)$$ ✓eq

When carbon dioxide is in excess or sodium hydroxide is limited, sodium hydrogen carbonate can be formed:✓Re

$$CO_2(g) + NaOH(aq) \rightarrow NaHCO_3(aq)$$ ✓eq

**(c) Suitable applications, health and environmental implications and clear mitigation measures**

**Aluminium oxide**

Aluminium oxide is used in abrasives, ceramics, furnace linings and aluminium extraction because it is hard and has a very high melting point✓Ea. Fine aluminium oxide dust can irritate the eyes and respiratory system.✓Ed Workers should use dust masks or N95 respirators, safety goggles and gloves, while good ventilation and dust extraction should be provided. Waste should be collected in sealed containers to prevent dust pollution.✓Em

**Chlorine**



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Chlorine is used for water purification, swimming-pool disinfection and manufacture of bleach because it forms hypochlorous acid, which kills microorganisms.✓Eg Chlorine gas is toxic when inhaled and can cause serious respiratory injury.✓Ed It should therefore be stored in secure, cool and well-ventilated areas, with chlorine gas detectors, mechanical ventilation and leak-control systems. Workers should use suitable protective clothing and respiratory protection. Chlorinated wastewater should be dechlorinated before discharge to protect aquatic life.✓Em

**Carbon dioxide**

Carbon dioxide is used in carbonated drinks, fire extinguishers, food preservation and refrigeration. ✓Eg High concentrations can displace oxygen and cause suffocation, especially in enclosed spaces.✓Ed Storage areas should have good ventilation and carbon dioxide detectors with alarms, while cylinders should be secured and kept away from excessive heat. Industries should reduce emissions through leak prevention, energy efficiency and carbon dioxide recovery or capture where practical.✓Em

ITEM 3

**1. Physical properties of the halogens**

The halogens are non-metallic elements found in Group VII of the periodic table. Fluorine and chlorine are gases at room temperature, bromine is a reddish-brown liquid, while iodine is a dark grey solid. Down the group, the atomic size and number of electrons increase, resulting in stronger London dispersion forces between the molecules. Therefore, the melting and boiling points generally increase down the group.✓Pe

The halogens exist as diatomic molecules, for example fluorine exists as $F_2$ and chlorine as $Cl_2$. They are poor conductors of electricity because they contain no mobile charged particles.✓Pe

The bond energy generally decreases from chlorine to iodine because the atoms become larger and the bonding electrons are further from the nuclei. The fluorine-fluorine bond is unusually weak because the small fluorine atoms have strong electron-electron repulsion between their bonding electrons. Therefore, the chlorine-chlorine bond has the highest bond energy among the four given halogens and requires more energy to break than the bromine-bromine and iodine-iodine bonds.✓Pe

**2. Physical properties of Group II carbonates**

Group II carbonates are ionic compounds consisting of Group II metal ions and carbonate ions. They are generally white or coloured crystalline solids with high melting points because strong electrostatic attractions exist between the oppositely charged ions.✓Pe

Their solubility generally increases down Group II. Magnesium carbonate is sparingly soluble, while the carbonates of calcium, strontium and barium become progressively more soluble.



The change in solubility occurs because, down the group, the attraction between the larger metal ions and carbonate ions becomes weaker, making it easier for the ionic lattice to separate and dissolve.✓Pe

**Chlorine and water**

Chlorine reacts reversibly with water because it undergoes **disproportionation**, where some chlorine is reduced to chloride while some is oxidised to chlorine in hypochlorous acid.✓Re

$$Cl_2(g) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$$ ✓eq

**Bromine and water**

Bromine also reacts reversibly with water by disproportioration, producing hydrobromic acid and hypobromous acid.

$$Br_2(l) + H_2O(l) \rightleftharpoons HBr(aq) + HBrO(aq)$$

**Iodine and water**

Iodine reacts only very slightly with water because it is less reactive and its iodine-iodine bond is relatively weak but iodine has a lower tendency to undergo the reaction with water under these conditions. The equilibrium lies strongly towards iodine and water, so very little reaction occurs.✓Re

$$I_2(s) + H_2O(l) \rightleftharpoons HI(aq) + HIO(aq)$$ ✓eq

**Fluorine and water**

Fluorine reacts very vigorously with water because fluorine is the most electronegative and strongest oxidising halogen. It oxidises water, producing hydrogen fluoride and oxygen.✓Re

$$2F_2(g) + 2H_2O(l) \rightarrow 4HF(aq) + O_2(g)$$ ✓eq

**Behaviour of halogens with alkalis**

**Chlorine with cold dilute sodium hydroxide**

Chlorine reacts with cold dilute sodium hydroxide by disproportionation. Chlorine is simultaneously reduced to chloride and oxidised to hypochlorite.✓Re

$$Cl_2(g) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$$ ✓eq

**Chlorine with hot concentrated sodium hydroxide**

When hot concentrated sodium hydroxide is used, chlorine undergoes further disproportionation to form sodium chloride and sodium chlorate.✓Re

$$3Cl_2(g) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$$ ✓eq

**3. Solubility of Group II carbonates in hydrochloric acid**



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Group II carbonates react with hydrochloric acid because carbonates contain carbonate ions, which react with hydrogen ions from the acid to form carbon dioxide and water.✓Re

For example, magnesium carbonate reacts with hydrochloric acid as follows:

$$MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)$$ ✓eq

Calcium carbonate reacts similarly:

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

The reaction occurs because hydrogen ions react with carbonate ions. Carbonic acid is initially formed but rapidly decomposes into carbon dioxide and water.

$$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$$

The visible effervescence observed is therefore due to the formation of carbon dioxide gas.

c) Chlorine is used in water treatment, swimming pools and cleaning agents because it reacts with water to form hypochlorous acid, which is a strong oxidising agent that kills microorganisms.✓Eo However, chlorine is **toxic and irritating** when present in high concentrations, and excessive chlorine discharged into water can harm aquatic organisms✓Ed. Therefore, chlorine products should be used at the recommended concentration, stored in closed containers and handled in well-ventilated areas. Workers should wear chemical-resistant gloves and safety goggles.✓Em

**ITEM 6**

**(a)**

**Sodium chloride**

Sodium chloride is a solid with high melting and boiling points.

Sodium chloride has a giant ionic structure with strong ionic bonds between the sodium and chloride ions in the lattice, making it a solid. The ionic bonds require a lot of heat energy to break and hence the high melting point.✓Pe

**Aluminium chloride**

Aluminium chloride is a solid, however, with low melting and boiling points.

Aluminium chloride is supposed to be ionic, but due to the small ionic radius and high ionic charge, the aluminium ion has a high charge density and high polarising power and hence greatly polarises the chloride ions, making it covalent partly. Its molecules interact via weak van der Waals forces, which are, however, strong enough to make it a solid but can easily be broken under low amounts of heat. ✓Pe

**Tin(IV) chloride**



Tin(IV) chloride has a simple molecular structure, as well as chlorine. Their molecules interact via weak van der Waals forces of attraction and hence require less heat energy to break, hence low melting point.

Tin(IV) chloride, however, has a higher molecular mass and hence stronger van der Waals forces of attraction. This makes it a liquid since its molecules are more closely and strongly attracted to each other and hence has a higher melting point/boiling point than chlorine. ✓Pe

**Reactions with water**

Sodium chloride dissociates in water to form sodium ions and chloride ions. This is because it is a salt formed from a strong acid and strong base and hence it does not undergo hydrolysis. ✓Re

$$NaCl(aq) \rightarrow Na^+(aq) + Cl^-(aq)$$ ✓Eq

Aluminium chloride undergoes hydrolysis in water to form aluminium hydroxide and hydrochloric acid. This is due to the small ionic radius and high charge, hence high charge density and high polarising power. It greatly polarises the water molecules to release hydrogen ions. ✓Ae

$$Al^{3+}(aq) + 3H_2O(l) \rightleftharpoons Al(OH)_3(s) + 3H^+(aq)$$ ✓Eq

Tin (IV) chloride undergoes hydrolysis to form tin (IV) oxide and hydrogen chloride. This is because tin has a vacant d-orbital which accommodates a lone pair of electrons from the oxygen atom of water to form coordinate bonds, and hence the reaction. ✓Re

$$SnCl_4(l) + 2H_2O(l) \rightleftharpoons SnO_2(s) + 4HCl(aq)$$ ✓Eq

Chlorine disproportionates in water to form hydrochloric acid and hypochlorous acid. ✓Re

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HOCl(aq)$$ ✓Eq

Aluminium chloride hydrolyses to form an acidic solution and hence reacts with cold dilute sodium hydroxide to form aluminium hydroxide and sodium chloride. ✓Ae

$$AlCl_3(aq) + 3NaOH(aq) \rightarrow Al(OH)_3(s) + 3NaCl(aq)$$ ✓Eq

Chlorine disproportionates in cold dilute sodium hydroxide solution to form sodium chloride, sodium chlorate(I) and water. ✓Re

$$Cl_2(aq) + 2OH^-(aq) \rightarrow Cl^-(aq) + ClO^-(aq) + H_2O(l)$$ ✓Eq

**OR**

$$Cl_2(aq) + 2NaOH(aq) \rightarrow NaCl(aq) + NaOCl(aq) + H_2O(l)$$

**(b) Applications**



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Chlorine dissolves in water to form hypochlorous acid and hydrochloric acid. This lowers the pH, making it unfavourable for multiplication of microorganisms, and hypochlorous acid also destroys bacterial cell walls and oxidises their nucleic acids, preventing them from reproducing. ✓Ea

Chlorine, being a strong oxidising agent, oxidises compounds produced by algae and bacteria that would have caused unpleasant smell and taste.

Chlorine also oxidises dissolved substances such as iron(II) ions and manganese(II) ions, and these can hence be easily removed during filtration.

**Danger**

Long-term exposure to high concentrations of chlorine causes cancers like bladder cancer, liver damage and kidney damage. ✓Ed

Too much chlorine used in water at times causes irritation of the eyes and the respiratory system, resulting in severe coughing and difficulty in breathing.

High chlorine levels/concentrations corrode metal pipes, pumps and water tanks, and this shortens the equipment lifespan and releases metals into water, making it unsafe for human consumption/use.

**Mitigation**

Use alternative water-treatment methods like coagulation, filtration or sedimentation, which do not use poisonous chemicals while greatly improving water quality and hence avoiding diseases associated with chlorine. ✓Em

The amount of chlorine must be checked and controlled regularly to ensure that the appropriate levels are used in water before its supply for human consumption, and pH must be maintained in the optimum range (6.5–7.5).

## ITEM 7

**(a)**

Sodium oxide, magnesium oxide and aluminium oxide have giant ionic structures with strong ionic bonds between the metal ions and the oxide ions, which require high amounts of heat to break. ✓Pe

From sodium oxide to magnesium oxide, the heat required increases due to increase in magnitude of charge of the cation, hence forming stronger ionic bonds. ✓Pe

Aluminium oxide has relatively lower heat energy required/melting point due to the small ionic radius and high charge of the aluminium ion, which gives it a high charge density and high polarising power and hence greatly tends to polarise the oxide ion, reducing the ionic character. ✓Pe

Silicon(IV) oxide has a giant covalent/molecular structure with strong covalent bonds between the silicon and oxygen atoms, which are continuously bonded to each other. The bonds hence require a lot of heat to break, giving it a high melting point. ✓Pe



Sulphur trioxide has a simple molecular structure in which its molecules interact via weak van der Waals forces of attraction, which are easily broken under low heat energy, and hence it easily fumes since its particles are far apart, making it a gas. ✓Pe

**(b) Reactions**

Sodium oxide is highly ionic and hence reacts with cold water vigorously to form sodium hydroxide. ✓Re

$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$ ✓Eq

Magnesium oxide is also ionic, but the magnesium ion has a higher charge density, giving it a high lattice energy, hence dissolves slowly in cold water to form magnesium hydroxide. ✓Re

$MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(aq)$ ✓Eq

Sulphur trioxide is strongly acidic and polar hence highly reactive with cold water to form sulphuric acid. ✓Re

$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$ ✓Eq

Aluminium oxide, being amphoteric, reacts with hot concentrated sodium hydroxide to form sodium aluminate. ✓Re

$Al_2O_3(s) + 2OH^-(aq) + 3H_2O(l) \rightarrow 2Al(OH)_4^-(aq)$ ✓Eq

Sulphur trioxide, being acidic in nature, reacts with cold dilute sodium hydroxide to form sodium sulphate and water. ✓RE

$SO_3(g) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + H_2O(l)$ ✓Eq

**(c) Applications, Dangers and Mitigation**

| Oxide | Application(s) | Danger(s) | Mitigation |
|---|---|---|---|
| Sodium oxide, $Na_2O$ | Used in the manufacture of soda-lime glass since it is strongly basic and lowers the melting point of silica✓Ea | Reacts violently with moisture to form sodium hydroxide, which can cause skin and eye burns since it is corrosive. ✓Ed | Store in tightly closed containers away from moisture to avoid reacting. Wear gloves, goggles and protective clothing to prevent contact with the body. ✓Em |
| Magnesium oxide, MgO | Used in furnace and kiln linings because it withstands high temperatures and is | Dust particles cause irritation of the eyes and respiratory system, hence causing discomfort. | Use dust masks or maintain good workplace ventilation to avoid over- |



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| Oxide | Application(s) | Danger(s) | Mitigation |
|---|---|---|---|
| | chemically stable. Used as an electrical insulator since it is a poor conductor of electricity. | | accumulation/inhalation of dust. |
| Aluminium oxide, $Al_2O_3$ | It can be used in making sandpaper or grinding wheels because of its hardness, chemical stability and high melting point that make it hard and stable.<br><br>It is used in extraction of aluminium since it is amphoteric and reacts with concentrated alkalis to form an aluminate, which when heated forms the pure ore. | Dust particles irritate the eyes and respiratory system, causing discomfort/coughing.<br><br>Mining of $Al_2O_3$ (bauxite) ore leads to habitat destruction and soil erosion since soils are left bare due to clearing of land/space for mining. | Use dust masks to prevent inhalation of the dust particles.<br><br>Restore vegetation and rehabilitate mining sites to avoid leaving soil bare. |
| Silicon(IV) oxide, $SiO_2$ | Used in the manufacture of glass due to its giant covalent structure, which gives it hardness and the chemical and heat resistance required for glass. | Long-term inhalation of fine crystalline silica dust causes silicosis, a lung disease. | Install dust-extraction systems to reduce dust inhalation. |
| Sulphur trioxide, $SO_3$ | Used in the production of sulphuric acid. This is because it dissolves readily in concentrated sulphuric acid, forming | The acid formed is highly corrosive and causes severe burns.<br><br>It reacts with rainwater to | Wear chemical-resistant gloves to prevent body contact. |



| Oxide | Application(s) | Danger(s) | Mitigation |
|---|---|---|---|
| | oleum, which then dissolves in water to form sulphuric acid. | form acidic rain due to formation of $H_2SO_4$. | |

## NUMBER 5/6 SOLUTIONS

ITEM 1

**Ethene** has the lowest boiling point because it is a small, non-polar molecule. Its molecules are held together only by weak **London dispersion forces**. Therefore, little energy is required to overcome the intermolecular forces and separate the molecules. **Ethanol** has a much higher boiling point because it contains an **–OH group**. The –OH groups of neighbouring ethanol molecules form **hydrogen bonds**, which are stronger than London forces. More energy is therefore required to separate the molecules.✓Pe

**Propan-2-ol** also forms hydrogen bonds because it contains an –OH group. It has a slightly larger molecular size than ethanol, giving it stronger London dispersion forces in addition to hydrogen bonding. Consequently, its boiling point is slightly higher than that of ethanol. **Propanoic acid** has the highest boiling point because its **–COOH groups form strong hydrogen bonds**. Two carboxylic acid molecules can form hydrogen-bonded dimers: A large amount of energy is therefore required to separate the molecules.✓Pe

Water is a **polar solvent**, and substances that can form strong interactions, especially hydrogen bonds, with water tend to be more soluble.

**Ethene** is only slightly soluble in water because it is non-polar and cannot form hydrogen bonds with water. **Ethanol** is completely miscible with water because its polar **–OH group forms hydrogen bonds with water molecules**.✓Pe

**Propan-2-ol** is also highly soluble/miscible in water because its –OH group forms hydrogen bonds with water. However, its larger hydrocarbon portion makes the molecule somewhat more non-polar than ethanol.✓Pe

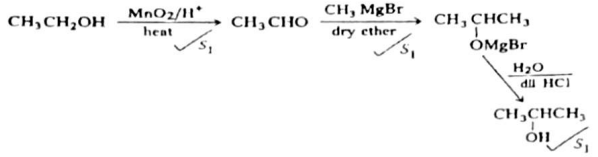
**Propanoic acid** is highly soluble in water because its **–COOH group forms hydrogen bonds** with water. It also partially ionises in water:✓Pe

b)

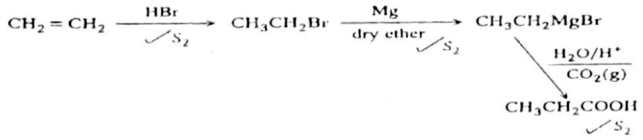


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**(1) Propan - 2 - ol from ethanol**

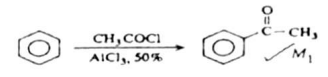


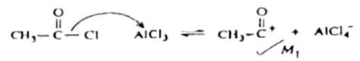

**(1) Propanoic acid from ethene**



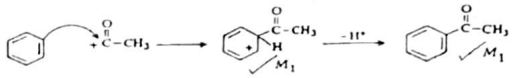

c)

iii)



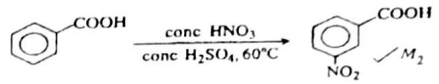



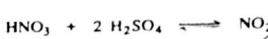

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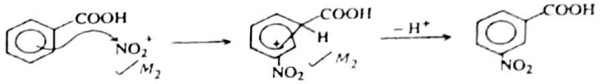





iv)




Phenylethanone is used as a **fragrance ingredient, solvent and intermediate in pharmaceutical and organic synthesis**✓Ea. Exposure can irritate the **eyes, skin and respiratory system**, while ethanoyl chloride and aluminium chloride used in the process are corrosive and reactive✓Ed. Workers should therefore wear **chemical-resistant gloves, safety goggles, laboratory coats and closed shoes**, while the reaction should be carried out in a **fume cupboard or sealed reactor with an extraction system**. Reagents should be added slowly while controlling the temperature, and chemical residues should be collected and treated rather than released into drains or water bodies.✓Em

3-Nitrobenzoic acid is used mainly as an **intermediate in pharmaceutical, dye and fine-chemical manufacture**✓Ea. The main danger during production is the use of concentrated nitric and sulfuric acids, which can cause severe chemical burns and release harmful fumes. ✓Ed Since nitration is also exothermic, excessive heating can make the reaction difficult to control. Workers should **wear acid-resistant gloves, goggles, face shields, laboratory coats and closed footwear**, and the reaction should be performed in a **fume cupboard or closed reactor fitted with an exhaust system**. The temperature should be monitored continuously and the acids added gradually. Acidic wastewater should be **neutralised with a suitable alkali before disposal**, and solid and organic chemical waste should be collected separately for controlled treatment✓Em.

**Item 5**

**Reactants/raw materials**: propan-1-ol and ethanol, ethyne
**Products:** ethanol and benzoic acid.

**Ethyne** has the lowest boiling point because it is non-polar due to almost equal electronegativity between hydrogen and carbon. Its molecules therefore interact via weak van der Waals forces which easily break when low heat energy is used.✓Pe



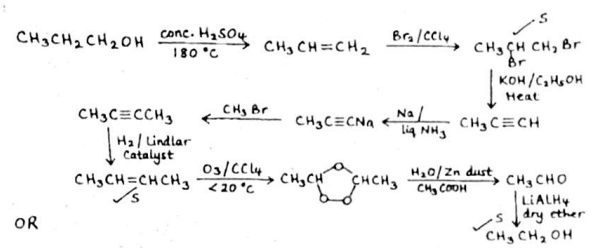
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**Ethanol and propan-1-ol** have high boiling points since they each possess a polar hydroxyl group and therefore their molecules interact via strong hydrogen bonds which require a lot of heat energy to break.✓Pe

The boiling point of **propan-1-ol** is higher than that of ethanol due to higher molecular weight and hence stronger van der Waals forces between the non-polar carbon chains, requiring higher amounts of heat energy to separate the molecules.✓Pe

**Benzoic acid** has the highest boiling point due to formation of strong hydrogen bonds between the polar hydroxyl groups and each two molecules interact via two hydrogen bonds, making the bonds many compared to those in alcohols and hence require higher amounts of heat energy to break.✓Pe

**b)**

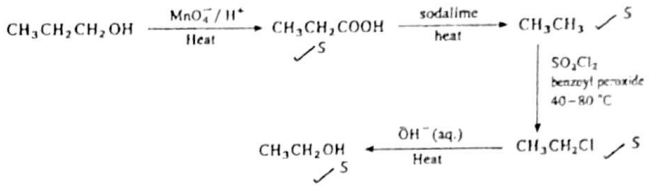


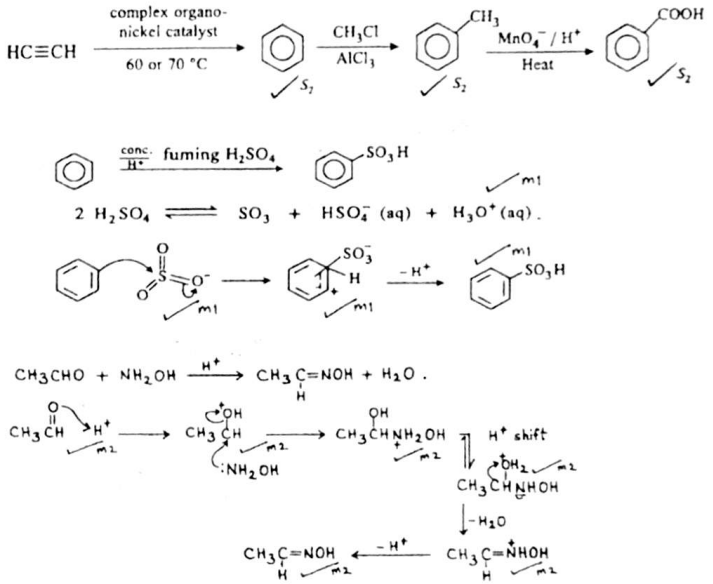

OR

OR

OR






**C)** Propan-1-ol is used as an industrial solvent. This is because it dissolves many organic substances due to the presence of the polar hydroxyl group and non-polar carbon chain, enabling it dissolve both polar and non-polar compounds.✓Ea



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**Danger:**

Highly flammable and can easily cause fire accidents resulting into loss of property and life.✓Ed

**Mitigation:**

Store away from heat and open flames to avoid catching fire and hence control any loses.✓Em

**Danger 2:**

Prolonged inhalation causes dizziness, headaches and irritation of the eyes and respiratory system making it poisonous to humans.✓Ed

**Mitigation 2:**

Wear personal protective gears like masks to control inhalation and reduce on the poisonous effects of irritation.✓Em

**Ethyne**

It is used for welding and cutting steel and other metals. This is because it burns to produce one of the hottest flames.✓Eo

**Danger:**

Extremely flammable. Cylinders can explode if mishandled resulting into accidents and hence loss of lives.✓Ed

**Mitigation:**

Avoid any leaks and store away from heat and open fire. Store cylinders upright and tightly closed to control escape of any gases and hence avoid the likely dangers.✓Em

**ITEM 6**

a) Ethene and Ethylethanoate evaporate easily because they have low boiling points. This is because they are largely non-polar and hence their molecules interact via weak van der Waal's forces of attraction which easily break for the liquid molecules to go into vapour phase.✓Pe

However, ethylethanoate requires a higher temperature than ethene due to a high molecular weight and hence stronger van der Waal's forces which require higher amounts of heat energy to break than those in ethene.✓Pe

Ethanol and propan-1-ol have higher boiling points and hence do not easily evaporate. Their molecules have polar hydroxyl groups and hence interact via strong hydrogen bonds which require a lot of heat energy to break.✓Pe

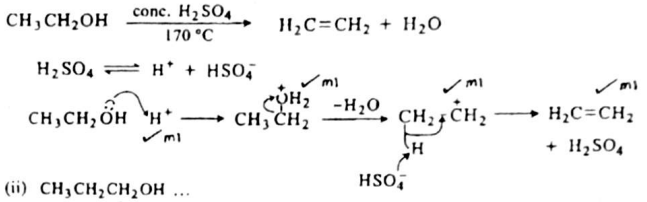
Propan-1-ol with a higher molecular weight requires more heat energy to separate its molecules than ethanol due to stronger van der Waal's forces between the non-polar carbon chain which require more heat energy to break than in ethanol.✓Pe

Ethanoic acid requires very high temperature to evaporate/boil than ethanol and propan-1-ol. Its molecules interact via two hydrogen bonds for every two molecules yet those in alcohols only form



one hydrogen bond. The high number of hydrogen bonds require a lot of heat energy to break and hence very high vaporisation temperature.✓Pe

b)

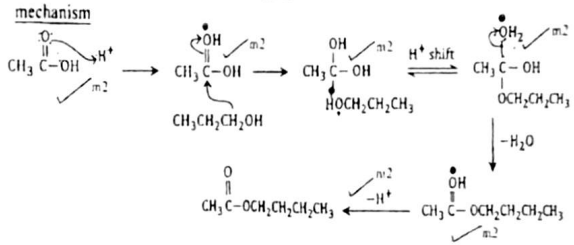


(ii) $CH_3CH_2CH_2OH$ ...

ii)

(ii) $CH_3CH_2CH_2OH + CH_3COOH \xrightarrow[\text{Heat}]{H^+} CH_3COOCH_2CH_2CH_3 + H_2O$

mechanism





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(a) $H_2C=CH_2 \xrightarrow[\text{Heat}]{H^+ \mid H_2O} CH_3CH_2OH \underset{\text{Heat}}{\overset{CH_3COOH / H^+}{\rightleftharpoons}} CH_3COOCH_2CH_3 + H_2O$. $S_1$ $S_1$

(b) $H_2C=CH_2 + Br_2 \xrightarrow{CCl_4} CH_2BrCH_2Br \xrightarrow[\text{Heat}]{\text{Ex(alc.) } KOH \mid C_2H_5OH} HC\equiv CH \xrightarrow[NH_3]{Na \mid Liq} HC\equiv CNa \xrightarrow{CH_3Br} CH_3C\equiv CH \xrightarrow[HgSO_4,\ 60^\circ C]{H_2O \mid conc\ H_2SO_4} CH_3COCH_3$

$S_2$ $S_2$ $S_2$ $S_2$ $S_2$

c)

Ethylethanoate can be used as a fragrance additive for soaps and cosmetics. This is because it is an ester with a sweet, fruity smell and hence gives the soap and cosmetics a pleasant smell.✓Ea

**Dangers**

**Highly flammable and explosive** since its vapours ignite easily when exposed to heat or sparks, causing flames which cause fires destroying property and lives.✓Ed mitigated by Storing in tightly closed containers away from heat and open flames to avoid any possibility of catching fire, and keep fire extinguishers nearby and functional to control any fire outbreaks.✓Em

**Danger 2**

Liquid or vapour irritates the eyes and skin, causing discomfort and cracking of the skin due to dehydration of body cells.✓Ed mitigated by wearing of protective equipment like safety goggles to prevent inhalation and chemical-resistant gloves as well as lab coats to control any contact with the skin.✓Em

**Application**

**Ethanol and propan-1-ol** can be used as cleaning solvents for industrial and laboratory use because they dissolve a range of organic compounds since they possess a polar hydroxyl group forming hydrogen bonds with polar solutes and the non-polar carbon chain forming van der Waals forces with non-polar solutes.✓Ea

**Danger - refer to the above dangers and mitigations of ethyl ethanoate in addition to:**

**Ethanol**



Excessive consumption of ethanol found in alcoholic drinks leads to dependence, impaired judgement, liver diseases and social problems as one's judgement is poor.✓Ed **Mitigation:** Promote responsible consumption through public education and legal regulation of alcohol sales as well as rehabilitation programmes for alcohol addicts to reduce over dependence/consumption.✓Em

**Propan-1-ol** is highly toxic and can cause poisoning when ingested or inhaled in large amounts, and this can result in death.✓Ed **Mitigation:** Labelling of containers clearly and storing securely away from food and beverages to avoid any likely poisoning or ingestion.✓Em

## PRACTICAL /PAPER 2

## PRACTICAL ITEMS

### ITEM ONE

Ammonia solution is used in the manufacture of household and industrial cleaning products. Its ionisation in water affects the concentration of hydroxide ions produced and is therefore important when assessing ammonia solutions intended for such products.

A local company uses aqueous ammonia to manufacture cleaning products. During quality control, the company obtained a new batch of ammonia solution and needed to assess its ionisation behaviour before using it. The company considers the sample suitable when its experimentally determined enthalpy of ionisation is between +4.5 and +8.0 kJ $mol^{-1}$.

Since the enthalpy of ionisation of ammonia cannot easily be measured directly using the available laboratory equipment, a company chemist recommended using Hess's law. Senior Five learners from a nearby school were given a sample from the new batch and asked to investigate it. The chemist advised them to determine the enthalpy changes when equal volumes of ammonia solution and sodium hydroxide solution of the same concentration react separately with portions of hydrochloric acid under identical conditions. The temperature of each mixture was to be measured at suitable time intervals.

The following equations were provided:

$NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

The ionisation of ammonia in water is represented by:

$NH_3(aq) + H_2O(l) \rightarrow NH_4^+(aq) + OH^-(aq)$

The learners were, however, unable to complete the investigation and have therefore approached you for help.

The following information was also provided:

- Specific heat capacity of the solutions = 4.2 J $g^{-1}$ $K^{-1}$
- Density of the solutions = 1.0 g $cm^{-3}$

You are provided with:



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- FA1, which is 30 $cm^3$ of 2.0 M ammonia solution.
- FA2, which is 30 $cm^3$ of 2.0 M sodium hydroxide solution.
- FA3, which is 100 $cm^3$ of 2.0 M hydrochloric acid.

Task

Design and carry out an investigation to help the learners and use your findings to advise the company on the suitability of the sample for use in its cleaning products. Present your response in the form of a scientific report.

## RESPONSES

**Aim**

An investigation to determine the enthalpy of ionisation of ammonia by adding increasing volumes of hydrochloric acid✓A to fixed volumes of ammonia solution and sodium hydroxide solution✓A separately and measuring the maximum temperature attained after each addition✓A, in order to assess the quality and suitability of the ammonia sample for use in cleaning products. ✓A

**Hypothesis**

The enthalpy of ionisation of the ammonia sample obtained by adding increasing volumes of hydrochloric ✓H to fixed volumes of ammonia solution and sodium hydroxide solution separately and measuring the maximum temperature✓H attained after each addition will lie within the reference range of +4.5 to +6.5 kJ $mol^{-1}$,✓H indicating that the sample is suitable for use in cleaning products.

**Variables**

**Independent variable**: Volume of FA3 added, Nature of alkali used✓V

**Dependent variable**: Maximum temperature attained by the reaction mixture. ✓V

**Controlled variables**: Volume of the alkalis used was fixed at 25.0$cm^3$ for each✓V

**Procedure**

(a) 25.0 $cm^3$ ✓Pc of FA1 was measured using a pipette/measuring cylinder✓Pr and transferred into a clean polystyrene cup. ✓Pr
(b) The initial temperature of FA1 was measured using a thermometer✓Pr and recorded.
(c) 5.0 $cm^3$ ✓Pc portions of FA3 was measured and added to FA1. The mixture was stirred and the maximum temperature attained was recorded after every addition
(d) The procedure (a) to (c) were repeated using fresh 25.0 $cm^3$ portions of FA2 instead of FA1 under identical conditions. ✓Pc
(e) The results were tabulated.

**Mitigated risks**

- Skin and eye irritation from hydrochloric acid and sodium hydroxide since they are corrosive: This may occur when the solutions splash or spill onto the skin or into the eyes during measuring, transferring or mixing. The risk was mitigated by wearing protective gloves, a laboratory coat and safety goggles, which provide a barrier that prevents the solutions from coming into direct contact with the skin and eyes✓Rm
- Irritation of the respiratory system from ammonia fumes: This may occur when ammonia vapour escapes from the solution and is inhaled during measuring, transferring or mixing. The



risk was mitigated by carrying by wearing gas masks to prevent direct inhalation of gaseous fumes✓Rm
- Cuts and injuries from broken glass apparatus: This may occur when a thermometer or other glass apparatus is dropped, knocked over or broken during handling, producing sharp pieces of glass that may cut the hands or feet. The risk was mitigated by wearing personal protective equipment such as closed shoes, protective gloves and a lab coat to prevent direct contact with broken gass✓Rm

**Data presentation**

Initial temperature of FA1 = 25.5 °C

Initial temperature of FA2 = 25.5 °C

Volume of FA1/FA2 used in each experiment = 25.0 $cm^3$

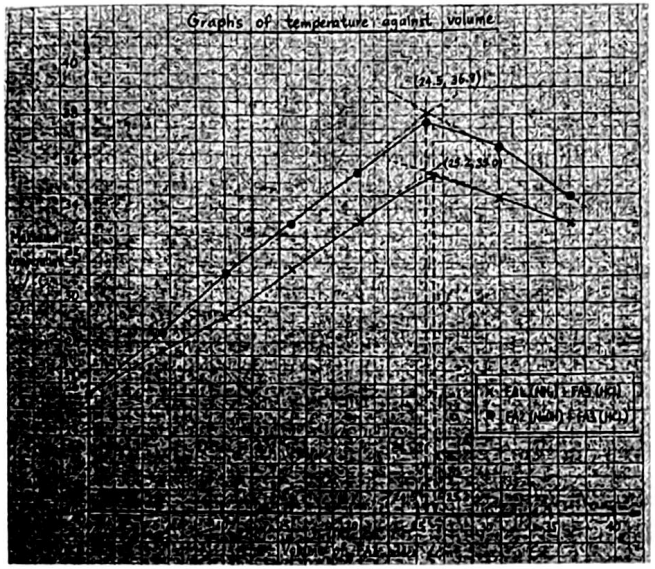
| Volume of FA3 added / $cm^3$ | 0.0 | 5.0 | 10.0 | 15.0 | 20.0 | 25.0 | 30.0 | 35.0 |
|---|---|---|---|---|---|---|---|---|
| Maximum temperature of FA1 ($NH_3$) + FA3 (HCl) / °C | 25.5 | 27.5 | 29.0 | 31.0 | 33.0 | 35.0 | 34.0 | 33.0 |
| Maximum temperature of FA2 (NaOH) + FA3 (HCl) / °C | 25.5 | 28.0 | 31.0 | 33.0 | 35.0 | 36.5 | 36.0 | 34.0 |

✓Dr ✓Dt ✓Dc ✓Dc

Graphical analysis



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**Analysis of data (Graphs**

- Correctly used scale✓Da
- Correct plots ✓Da
- Extrapolation✓Da
- Labelled axes with units✓Da
- Shapes of graphs✓Da

Initial temperature of both solutions = 25.5 °C

For FA1 ($NH_3$) + FA3 (HCl): $\Delta T_1$ = 35.0 − 25.5 = 9.5 °C✓Da

For FA2 (NaOH) + FA3 (HCl): $\Delta T_2$ = 36.9 − 25.5 = 11.4 °C✓Da

Total volume of solution = 25.2 + 25.0 = 50.2 $cm^3$

Total volume of solution = 24.5 + 25.0 = 49.5 $cm^3$

Assuming density = 1.0 g $cm^{-3}$, mass of solution = 50.0 g.

Specific heat capacity of solution = 4.2 J $g^{-1}$ $°C^{-1}$.

Determination of enthalpy change for FA1 + FA3

$NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$



$Q = mc\Delta T$; (50.2 × 4.2 × 9.5) = 2002.98J

Moles of $NH_3$: $\frac{2.0}{25.0 \times 1000}$ = 0.050 moles

Molar enthalpy of reaction in kJ; $\frac{2002.98}{0.050 \times 1000}$ = −40.1kJ $mol^{-1}$ ✓Da

The negative value shows that the reaction between ammonia and hydrochloric acid is exothermic.

Determination of enthalpy change for FA2 + FA3

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

$Q = mc\Delta T$; (49.5 × 4.2 × 11.4) = 2370

Moles of NaOH, $\frac{2.0}{25.0 \times 1000}$ = 0.050 moles

Molar enthalpy of reaction in kJ; $\frac{2370}{0.050 \times 1000}$ = −47.4kJ $mol^{-1}$ ✓Da

The negative value shows that neutralisation of hydrochloric acid by sodium hydroxide is exothermic.

Determination of enthalpy of ionisation of ammonia

Required reaction:

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

Reaction I: $NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$. $\Delta H_1$ = −40.1kJ $mol^{-1}$

Reaction II: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ $\Delta H_2$ = −47.4 kJ $mol^{-1}$

Reversing Reaction II gives: $H_2O(l) \rightarrow H^+(aq) + OH^-(aq)$. $\Delta H$ = +47.4 kJ $mol^{-1}$

By Hess's law: $\Delta H_{ion}$ = −40.1 + 47.4 = +7.3kJ $mol^{-1}$ ✓Da

Interpretation: The positive value indicates that the ionisation of ammonia is endothermic. ✓D2

**Conclusion**

The enthalpy of ionisation of the ammonia sample was determined experimentally using Hess's law as approximately +7.3 kJ $mol^{-1}$. The positive value confirms that ionisation of ammonia in water is an endothermic process. ✓C

Since +7.3 kJ $mol^{-1}$ is within company's acceptable range of +4.5 to +8.0 kJ $mol^{-1}$, the experimental results do support the hypothesis, and the sample meets the stated quality requirement.

**Recommendation**

The ammonia sample should be approved for use in the manufacture of the company's cleaning products because its experimentally determined enthalpy of ionisation falls within the specified range. ✓R

ITEM 2

Qualitative analysis of inorganic ions is important because it helps identify substances that affect water quality and guides decisions on the safe use of water for domestic purposes.

In a growing residential area, residents have reported persistent stains on cooking utensils and a gritty deposit forming in stored water containers. Community leaders suspect that runoff from a



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nearby construction site is washing inorganic materials into local water sources, especially during periods of heavy rainfall.

To investigate the problem, a sample of the suspected material, labelled Substance R, was collected and submitted to your school's chemistry laboratory since the community lacks the facilities and expertise required to carry out the investigation.

A laboratory technician observed that Substance R was only partially soluble in water and advised that qualitative analysis of the soluble and insoluble portions could be used to identify the two cations and two anions suspected to be responsible for the staining and sediment formation.

Your chemistry teacher has chosen you to carry out this investigation since you have knowledge on qualitative analysis.You are provided with: Substance R

Task, As a chemistry learner:

(a) Plan and carryout an investigation to help the residents.
(b) Conclude on the identity of the ions present in Substance R and advise the community on the likely effects of any of the identified ions on household water use.

| TEST | OBSERVATION | DEDUCTION |
|---|---|---|
| (a) A small amount of substance Z was heated strongly in a dry test tube until there was no further change✓$^{Pp}$ | The white solid decomposes to form a residue which is yellow when hot and white when cold✓$^{op}$<br>Colourless condensate turns anhydrous copper(II) sulphate blue. ✓$^{op}$<br>White fumes turn blue litmus paper red✓$^{op}$<br>Gas turns lime water milky ✓$^{op}$<br>Gas turns acidified potassium dichromate from orange to green✓$^{op}$ | Non-transition metal cations present or $Zn^{2+}$, $Mg^{2+}$, $Al^{3+}$ present✓$^{Dp}$<br>Water of crystallisation/Hydrated salt✓$^{Dp}$<br>Acidic gas✓$^{Dp}$<br>$CO_2/CO_3^{2-}$ present✓$^{Dp}$<br>$SO_2/SO_4^{2-}$ present✓$^{Dp}$ |
| (b) Two spatula endfuls of Z were shaken with distilled water and the mixture was filtered. ✓$^{Pp}$ Both the residue and filtrate were kept. The filtrate was divided into six portions | Partially soluble to form a white suspension<br>A colourless filtrate formed✓$^{op}$<br>white residue was obtained. ✓$^{op}$ | Non-transition metal cation in both the filtrate and residue✓$^{Dp}$ |
| (i) To the first portion, sodium hydroxide solution was added dropwise until in excess✓$^{Ps}$ | A white precipitate formed insoluble in excess sodium hydroxide. ✓$^{Oe}$ | $Ba^{2+}$, $Ca^{2+}$, $Mg^{2+}$ suspected present✓$^{Ds}$ |
| (ii) To the second portion, ammonia solution was added dropwise until in excess✓$^{Ps}$ | White precipitate insoluble in excess✓$^{Os}$ | $Ba^{2+}$, $Mg^{2+}$ suspected present✓$^{Ds}$ |



| | | |
|---|---|---|
| (iii) To the third portion dilute sulphuric acid or sodium sulphate was added✓$^{Ps}$ | No observable change✓$^{Os}$ | $Ba^{2+}$ absent ✓$^{Ds}$<br>$Mg^{2+}$ suspected present✓$^{Ds}$ |
| (iv) To the fourth portion, added solid ammonium chloride followed by disodium hydrogen phosphate solution and excess ammonia solution✓$^{Ps}$<br>OR<br>Added solid ammonium chloride followed by ammonia solution | White precipitate insoluble in excess ammonia✓$^{Oc}$<br>OR<br>No observable change | $Mg^{2+}$ confirmed present✓$^{Dc}$ |
| (v) To the fifth portion, added lead (II) nitrate followed by dilute nitric ✓$^{Ps}$ | White precipitate formed insoluble in the acid✓$^{Os}$ | $SO_4^{2-}$, $Br^-$, $I^-$ present✓$^{Ds}$ |
| (vi) To the sixth portion, added barium nitrate solution followed by dilute nitric acid✓$^{Ps}$<br>OR<br>added barium chloride solution followed by dilute hydrochloric acid. | A white precipitate formed insoluble in excess✓$^{Oc}$ | $SO_4^{2-}$ present✓$^{Dc}$ |
| (vii) (c) The residue was washed with water and dissolved in dilute nitric acid. ✓$^{Pc}$<br>The resultant solution was divided into three portions | Bubbles of a colourless gas which turns blue litmus paper red and limewater milky✓$^{Oc}$<br>Colourless solution formed ✓$^{Oe}$ | $CO_2/CO_3^{2-}$ confirmed present✓$^{Dc}$<br>$Zn^{2+}$, $Al^{3+}$, $Pb^{2+}$, $Sn^{2+}$ present✓$^{Ds}$ |
| (i) To the first portion, sodium hydroxide solution was added dropwise until in excess✓$^{Ps}$ | A white precipitate formed soluble in excess to form a colourless✓$^{Os}$ | $Zn^{2+}$, $Al^{3+}$, $Pb^{2+}$, $Sn^{2+}$ present✓$^{Ds}$ |
| (ii) To the second portion, ammonia solution was added dropwise until in excess✓$^{Ps}$ | A white precipitate formed soluble in excess to form a colourless✓$^{Os}$ | $Zn^{2+}$ present✓$^{Ds}$ |
| (iii) To the fourth portion, added solid ammonium chloride followed by disodium hydrogen phosphate solution and excess ammonia solution✓$^{Ps}$ | White precipitate soluble in excess ammonia to form a colourless solution✓$^{Oc}$ | $Zn^{2+}$ present ✓$^{Dc}$ |

(b) Conclusion on the identity of the ions present and advice to the community

Cations: $Mg^{2+}$✓$^{C}$ and $Zn^{2+}$✓$^{C}$

Anions: $SO_4^{2-}$ ✓$^{C}$ and $CO_3^{2-}$✓$^{C}$



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Advise to the community

- ✓ Magnesium ions ($Mg^{2+}$) contribute to temporary or permanent water hardness, causing poor soap lathering, increased soap consumption and stains on cooking utensils. ✓a
- ✓ Carbonate ions ($CO_3^{2-}$) react with hardness ions to form insoluble carbonates that produce gritty deposits and scale in storage containers, kettles and water pipes.
- ✓ Sulphate ions ($SO_4^{2-}$) are generally safe at low concentrations but excessive amounts may affect the taste of water and can have a mild laxative effect.
- ✓ Zinc ions ($Zn^{2+}$) are essential in trace amounts but elevated concentrations may impart an undesirable metallic taste and reduce the suitability of water for domestic use.
- ✓ The community should reduce runoff from the nearby construction site by improving drainage and erosion control measures, protect water sources from contamination, and treat the water through sedimentation, filtration and, where necessary, water softening before domestic use.✓a (One danger and one mitigation)

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